

7.14.3 Process: GATE CSE 2002 | Question: 2.21



Which combination of the following features will suffice to characterize an OS as a multi-programmed OS?

- a. More than one program may be loaded into main memory at the same time for execution
- b. If a program waits for certain events such as I/O, another program is immediately scheduled for execution
- c. If the execution of a program terminates, another program is immediately scheduled for execution.

- A. (a)
- B. (a) and (b)
- C. (a) and (c)
- D. (a), (b) and (c)

gatecse-2002 operating-system normal process

7.14.4 Process: GATE CSE 2023 | Question: 12



Which one or more of the following need to be saved on a context switch from one thread (T1) of a process to another thread (T2) of the same process?

- A. Page table base register
- B. Stack pointer
- C. Program counter
- D. General purpose registers

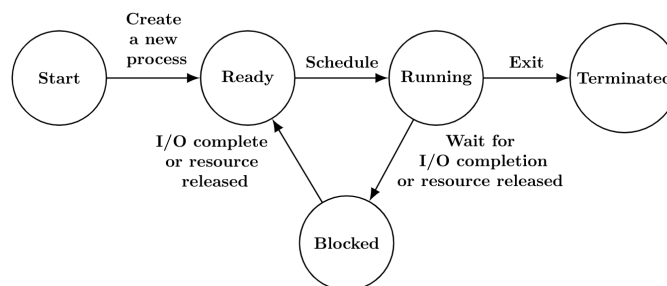
gatecse-2023 operating-system process multiple-selects 1-mark

7.14.5 Process: GATE IT 2006 | Question: 13



The process state transition diagram of an operating system is as given below.

Which of the following must be FALSE about the above operating system?



- A. It is a multiprogrammed operating system
- B. It uses preemptive scheduling
- C. It uses non-preemptive scheduling
- D. It is a multi-user operating system

gateit-2006 operating-system normal process

7.15

Process Scheduling (47)

7.15.1 Process Scheduling: GATE CSE 1988 | Question: 2xa



State any undesirable characteristic of the following criteria for measuring performance of an operating system:

Turn around time

gate1988 normal descriptive operating-system process-scheduling

7.15.2 Process Scheduling: GATE CSE 1988 | Question: 2xb



State any undesirable characteristic of the following criteria for measuring performance of an operating system:

Waiting time

gate1988 normal descriptive operating-system process-scheduling

Answer key

7.15.3 Process Scheduling: GATE CSE 1990 | Question: 1-vi



The highest-response ratio next scheduling policy favours _____ jobs, but it also limits the waiting time of _____ jobs.

gate1990 operating-system process-scheduling fill-in-the-blanks

Answer key

7.15.4 Process Scheduling: GATE CSE 1993 | Question: 7.10



Assume that the following jobs are to be executed on a single processor system

Job Id	CPU Burst Time
p	4
q	1
r	8
s	1
t	2

The jobs are assumed to have arrived at time 0^+ and in the order p, q, r, s, t . Calculate the departure time (completion time) for job p if scheduling is round robin with time slice 1

- A. 4 B. 10 C. 11 D. 12
E. None of the above

gate1993 operating-system process-scheduling normal

Answer key

7.15.5 Process Scheduling: GATE CSE 1995 | Question: 1.15



Which scheduling policy is most suitable for a time shared operating system?

- A. Shortest Job First B. Round Robin
C. First Come First Serve D. Elevator

gate1995 operating-system process-scheduling easy

Answer key

7.15.6 Process Scheduling: GATE CSE 1995 | Question: 2.6



The sequence _____ is an optimal non-preemptive scheduling sequence for the following jobs which leaves the CPU idle for _____ unit(s) of time.

Job	Arrival Time	Burst Time
1	0.0	9
2	0.6	5
3	1.0	1

- A. {3,2,1},1 B. {2,1,3},0 C. {3,2,1},0 D. {1,2,3},5

gate1995 operating-system process-scheduling normal

Answer key

7.15.7 Process Scheduling: GATE CSE 1996 | Question: 2.20, ISRO2008-15



Four jobs to be executed on a single processor system arrive at time 0 in the order A, B, C, D . Their burst CPU time requirements are 4, 1, 8, 1 time units respectively. The completion time of A under round robin scheduling with time slice of one time unit is

- A. 10 B. 4 C. 8 D. 9

gate1996 operating-system process-scheduling normal isro2008

Answer key

7.15.8 Process Scheduling: GATE CSE 1998 | Question: 2.17, UGCNET-Dec2012-III: 43



Consider n processes sharing the CPU in a round-robin fashion. Assuming that each process switch takes s seconds, what must be the quantum size q such that the overhead resulting from process switching is minimized but at the same time each process is guaranteed to get its turn at the CPU at least every t seconds?

- A. $q \leq \frac{t-ns}{n-1}$ B. $q \geq \frac{t-ns}{n-1}$
C. $q \leq \frac{t-ns}{n+1}$ D. $q \geq \frac{t-ns}{n+1}$

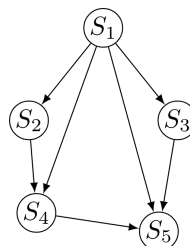
gate1998 operating-system process-scheduling normal ugcnetcse-dec2012-paper3

Answer key

7.15.9 Process Scheduling: GATE CSE 1998 | Question: 24



- a. Four jobs are waiting to be run. Their expected run times are 6, 3, 5 and x . In what order should they be run to minimize the average response time?
b. Write a concurrent program using `par` `begin-par` `end` to represent the precedence graph shown below.



gate1998 operating-system process-scheduling descriptive

Answer key

7.15.10 Process Scheduling: GATE CSE 1998 | Question: 7-b



In a computer system where the 'best-fit' algorithm is used for allocating 'jobs' to 'memory partitions', the following situation was encountered:

Partitions size in KB	4K 8K 20K 2K
Job sizes in KB	2K 14K 3K 6K 6K 10K 20K 2K
Time for execution	4 10 2 1 4 1 8 6

When will the 20K job complete?

gate1998 operating-system process-scheduling normal

Answer key

7.15.11 Process Scheduling: GATE CSE 2002 | Question: 1.22



Which of the following scheduling algorithms is non-preemptive?

- A. Round Robin
- C. Multilevel Queue Scheduling

- B. First-In First-Out
- D. Multilevel Queue Scheduling with Feedback

gatecse-2002 operating-system process-scheduling easy

Answer key

7.15.12 Process Scheduling: GATE CSE 2003 | Question: 77



A uni-processor computer system only has two processes, both of which alternate 10 ms CPU bursts with 90 ms I/O bursts. Both the processes were created at nearly the same time. The I/O of both processes can proceed in parallel. Which of the following scheduling strategies will result in the *least* CPU utilization (over a long period of time) for this system?

- A. First come first served scheduling
- B. Shortest remaining time first scheduling
- C. Static priority scheduling with different priorities for the two processes
- D. Round robin scheduling with a time quantum of 5 ms

gatecse-2003 operating-system process-scheduling normal

Answer key

7.15.13 Process Scheduling: GATE CSE 2004 | Question: 46



Consider the following set of processes, with the arrival times and the CPU-burst times gives in milliseconds.

Process	Arrival Time	Burst Time
P1	0	5
P2	1	3
P3	2	3
P4	4	1

What is the average turnaround time for these processes with the preemptive shortest remaining processing time first (SRPT) algorithm?

- A. 5.50
- B. 5.75
- C. 6.00
- D. 6.25

gatecse-2004 operating-system process-scheduling normal

Answer key

7.15.14 Process Scheduling: GATE CSE 2006 | Question: 06, ISRO2009-14



Consider three CPU-intensive processes, which require 10, 20 and 30 time units and arrive at times 0, 2 and 6, respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? Do not count the context switches at time zero and at the end.

- A. 1
- B. 2
- C. 3
- D. 4

gatecse-2006 operating-system process-scheduling normal isro2009

Answer key

7.15.15 Process Scheduling: GATE CSE 2006 | Question: 64



Consider three processes (process id 0, 1, 2 respectively) with compute time bursts 2, 4 and 8 time units. All processes arrive at time zero. Consider the longest remaining time first (LRTF) scheduling algorithm. In LRTF ties are broken by giving priority to the process with the lowest process id. The average turn around time is:

- A. 13 units
- B. 14 units
- C. 15 units
- D. 16 units

gatecse-2006 operating-system process-scheduling normal

Answer key

7.15.16 Process Scheduling: GATE CSE 2006 | Question: 65



Consider three processes, all arriving at time zero, with total execution time of 10, 20 and 30 units, respectively. Each process spends the first 20% of execution time doing I/O, the next 70% of time doing computation, and the last 10% of time doing I/O again. The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running process gets blocked on I/O or when the running process finishes its compute burst. Assume that all I/O operations can be overlapped as much as possible. For what percentage of time does the CPU remain idle?

- A. 0% B. 10.6% C. 30.0% D. 89.4%

gatecse-2006 operating-system process-scheduling normal

Answer key

7.15.17 Process Scheduling: GATE CSE 2007 | Question: 16



Group 1 contains some CPU scheduling algorithms and Group 2 contains some applications. Match entries in Group 1 to entries in Group 2.

Group I	Group II
(P) Gang Scheduling	(1) Guaranteed Scheduling
(Q) Rate Monotonic Scheduling	(2) Real-time Scheduling
(R) Fair Share Scheduling	(3) Thread Scheduling

- A. $P - 3; Q - 2; R - 1$ B. $P - 1; Q - 2; R - 3$ C. $P - 2; Q - 3; R - 1$ D. $P - 1; Q - 3; R - 2$

gatecse-2007 operating-system process-scheduling normal

Answer key

7.15.18 Process Scheduling: GATE CSE 2007 | Question: 55



An operating system used Shortest Remaining System Time first (SRT) process scheduling algorithm. Consider the arrival times and execution times for the following processes:

Process	Execution Time	Arrival Time
P1	20	0
P2	25	15
P3	10	30
P4	15	45

What is the total waiting time for process $P2$?

- A. 5 B. 15 C. 40 D. 55

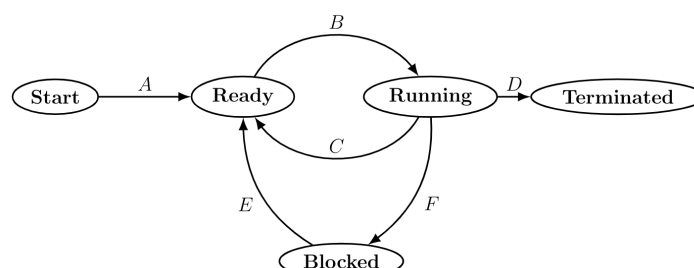
gatecse-2007 operating-system process-scheduling normal

Answer key

7.15.19 Process Scheduling: GATE CSE 2009 | Question: 32



In the following process state transition diagram for a uniprocessor system, assume that there are always some processes in the ready state:



Now consider the following statements:

- I. If a process makes a transition D , it would result in another process making transition A immediately.
- II. A process P_2 in blocked state can make transition E while another process P_1 is in running state.
- III. The OS uses preemptive scheduling.
- IV. The OS uses non-preemptive scheduling.

Which of the above statements are TRUE?

- A. I and II B. I and III C. II and III D. II and IV

gatecse-2009 operating-system process-scheduling normal

Answer key

7.15.20 Process Scheduling: GATE CSE 2010 | Question: 25



Which of the following statements are true?

- I. Shortest remaining time first scheduling may cause starvation
- II. Preemptive scheduling may cause starvation
- III. Round robin is better than FCFS in terms of response time

- A. I only B. I and III only C. II and III only D. I, II and III

gatecse-2010 operating-system process-scheduling easy

Answer key

7.15.21 Process Scheduling: GATE CSE 2011 | Question: 35



Consider the following table of arrival time and burst time for three processes P_0 , P_1 and P_2 .

Process	Arrival Time	Burst Time
P0	0 ms	9
P1	1 ms	4
P2	2 ms	9

The pre-emptive shortest job first scheduling algorithm is used. Scheduling is carried out only at arrival or completion of processes. What is the average waiting time for the three processes?

- A. 5.0 ms B. 4.33 ms C. 6.33 ms D. 7.33 ms

gatecse-2011 operating-system process-scheduling normal

Answer key

7.15.22 Process Scheduling: GATE CSE 2012 | Question: 31



Consider the 3 processes, P_1 , P_2 and P_3 shown in the table.

Process	Arrival Time	Time Units Required
P1	0	5
P2	1	7
P3	3	4

The completion order of the 3 processes under the policies FCFS and RR2 (round robin scheduling with CPU quantum of 2 time units) are

- A. **FCFS:** P_1, P_2, P_3 **RR2:** P_1, P_2, P_3
B. **FCFS:** P_1, P_3, P_2 **RR2:** P_1, P_3, P_2
C. **FCFS:** P_1, P_2, P_3 **RR2:** P_1, P_3, P_2
D. **FCFS:** P_1, P_3, P_2 **RR2:** P_1, P_2, P_3

gatecse-2012 operating-system process-scheduling normal

Answer key

7.15.23 Process Scheduling: GATE CSE 2013 | Question: 10



A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with zero (the lowest priority). The scheduler re-evaluates the process priorities every T time units and decides the next process to schedule. Which one of the following is **TRUE** if the processes have no I/O operations and all arrive at time zero?

- A. This algorithm is equivalent to the first-come-first-serve algorithm.
- B. This algorithm is equivalent to the round-robin algorithm.
- C. This algorithm is equivalent to the shortest-job-first algorithm.
- D. This algorithm is equivalent to the shortest-remaining-time-first algorithm.

gatecse-2013 operating-system process-scheduling normal

Answer key

7.15.24 Process Scheduling: GATE CSE 2014 Set 1 | Question: 32



Consider the following set of processes that need to be scheduled on a single CPU. All the times are given in milliseconds.

Process Name	Arrival Time	Execution Time
A	0	6
B	3	2
C	5	4
D	7	6
E	10	3

Using the *shortest remaining time first* scheduling algorithm, the average process turnaround time (in msec) is _____.

gatecse-2014-set1 operating-system process-scheduling numerical-answers normal

Answer key

7.15.25 Process Scheduling: GATE CSE 2014 Set 2 | Question: 32



Three processes A , B and C each execute a loop of 100 iterations. In each iteration of the loop, a process performs a single computation that requires t_c CPU milliseconds and then initiates a single I/O operation that lasts for t_{io} milliseconds. It is assumed that the computer where the processes execute has sufficient number of I/O devices and the OS of the computer assigns different I/O devices to each process. Also, the scheduling overhead of the OS is negligible. The processes have the following characteristics:

Process id	t_c	t_{io}
A	100 ms	500 ms
B	350 ms	500 ms
C	200 ms	500 ms

The processes A , B , and C are started at times 0, 5 and 10 milliseconds respectively, in a pure time sharing system (round robin scheduling) that uses a time slice of 50 milliseconds. The time in milliseconds at which process C would **complete** its first I/O operation is _____.

gatecse-2014-set2 operating-system process-scheduling numerical-answers normal

Answer key

7.15.26 Process Scheduling: GATE CSE 2014 Set 3 | Question: 32



An operating system uses *shortest remaining time first* scheduling algorithm for pre-emptive scheduling of processes. Consider the following set of processes with their arrival times and CPU burst times (in

milliseconds):

Process	Arrival Time	Burst Time
P1	0	12
P2	2	4
P3	3	6
P4	8	5

The average waiting time (in milliseconds) of the processes is _____.

gatecse-2014-set3 operating-system process-scheduling numerical-answers normal

Answer key

7.15.27 Process Scheduling: GATE CSE 2015 Set 1 | Question: 46



Consider a uniprocessor system executing three tasks T_1, T_2 and T_3 each of which is composed of an infinite sequence of jobs (or instances) which arrive periodically at intervals of 3, 7 and 20 milliseconds, respectively. The priority of each task is the inverse of its period, and the available tasks are scheduled in order of priority, which is the highest priority task scheduled first. Each instance of T_1, T_2 and T_3 requires an execution time of 1, 2 and 4 milliseconds, respectively. Given that all tasks initially arrive at the beginning of the 1st millisecond and task preemptions are allowed, the first instance of T_3 completes its execution at the end of _____ milliseconds.

gatecse-2015-set1 operating-system process-scheduling normal numerical-answers

Answer key

7.15.28 Process Scheduling: GATE CSE 2015 Set 3 | Question: 1



The maximum number of processes that can be in *Ready* state for a computer system with n CPUs is :

- A. n B. n^2 C. 2^n D. Independent of n

gatecse-2015-set3 operating-system process-scheduling easy

Answer key

7.15.29 Process Scheduling: GATE CSE 2015 Set 3 | Question: 34



For the processes listed in the following table, which of the following scheduling schemes will give the lowest average turnaround time?

Process	Arrival Time	Process Time
A	0	3
B	1	6
C	4	4
D	6	2

- A. First Come First Serve B. Non-preemptive Shortest job first
C. Shortest Remaining Time D. Round Robin with Quantum value two

gatecse-2015-set3 operating-system process-scheduling normal

Answer key

7.15.30 Process Scheduling: GATE CSE 2016 Set 1 | Question: 20



Consider an arbitrary set of CPU-bound processes with unequal CPU burst lengths submitted at the same time to a computer system. Which one of the following process scheduling algorithms would minimize the average waiting time in the ready queue?

- A. Shortest remaining time first
B. Round-robin with the time quantum less than the shortest CPU burst
C. Uniform random

D. Highest priority first with priority proportional to CPU burst length

gatecse-2016-set1 operating-system process-scheduling normal

Answer key

7.15.31 Process Scheduling: GATE CSE 2016 Set 2 | Question: 47



Consider the following processes, with the arrival time and the length of the CPU burst given in milliseconds. The scheduling algorithm used is preemptive shortest remaining-time first.

Process	Arrival Time	Burst Time
P_1	0	10
P_2	3	6
P_3	7	1
P_4	8	3

The average turn around time of these processes is _____ milliseconds.

gatecse-2016-set2 operating-system process-scheduling normal numerical-answers

Answer key

7.15.32 Process Scheduling: GATE CSE 2017 Set 1 | Question: 24



Consider the following CPU processes with arrival times (in milliseconds) and length of CPU bursts (in milliseconds) as given below:

Process	Arrival Time	Burst Time
P_1	0	7
P_2	3	3
P_3	5	5
P_4	6	2

If the pre-emptive shortest remaining time first scheduling algorithm is used to schedule the processes, then the average waiting time across all processes is _____ milliseconds.

gatecse-2017-set1 operating-system process-scheduling numerical-answers

Answer key

7.15.33 Process Scheduling: GATE CSE 2017 Set 2 | Question: 51



Consider the set of process with arrival time (in milliseconds), CPU burst time (in milliseconds) and priority (0 is the highest priority) shown below. None of the process have I/O burst time

Process	Arrival Time	Burst Time	Priority
P_1	0	11	2
P_2	5	28	0
P_3	12	2	3
P_4	2	10	1
P_5	9	16	4

The average waiting time (in milli seconds) of all the process using premtive priority scheduling algorithm is _____

gatecse-2017-set2 operating-system process-scheduling numerical-answers

Answer key

7.15.34 Process Scheduling: GATE CSE 2019 | Question: 41



Consider the following four processes with arrival times (in milliseconds) and their length of CPU bursts (in milliseconds) as shown below:

Process	P_1	P_2	P_3	P_4
Arrival Time	0	1	3	4
CPU burst time	3	1	3	Z

These processes are run on a single processor using preemptive Shortest Remaining Time First scheduling algorithm. If the average waiting time of the processes is 1 millisecond, then the value of Z is _____

gatecse-2019 numerical-answers operating-system process-scheduling 2-marks

Answer key

7.15.35 Process Scheduling: GATE CSE 2020 | Question: 12



Consider the following statements about process state transitions for a system using preemptive scheduling.

- I. A running process can move to ready state.
- II. A ready process can move to running state.
- III. A blocked process can move to running state.
- IV. A blocked process can move to ready state.

Which of the above statements are TRUE?

- A. I, II, and III only
- B. II and III only
- C. I, II, and IV only
- D. I, II, III and IV only

gatecse-2020 operating-system process-scheduling 1-mark easy

Answer key

7.15.36 Process Scheduling: GATE CSE 2020 | Question: 50



Consider the following set of processes, assumed to have arrived at time 0. Consider the CPU scheduling algorithms Shortest Job First (SJF) and Round Robin (RR). For RR, assume that the processes are scheduled in the order P_1, P_2, P_3, P_4 .

Processes	P_1	P_2	P_3	P_4
Burst time (in ms)	8	7	2	4

If the time quantum for RR is 4 ms, then the absolute value of the difference between the average turnaround times (in ms) of SJF and RR (round off to 2 decimal places) is _____

gatecse-2020 numerical-answers operating-system process-scheduling 2-marks

Answer key

7.15.37 Process Scheduling: GATE CSE 2021 Set 1 | Question: 25



Three processes arrive at time zero with CPU bursts of 16, 20 and 10 milliseconds. If the scheduler has prior knowledge about the length of the CPU bursts, the minimum achievable average waiting time for these three processes in a non-preemptive scheduler (rounded to nearest integer) is _____ milliseconds.

gatecse-2021-set1 operating-system process-scheduling numerical-answers 1-mark

Answer key

7.15.38 Process Scheduling: GATE CSE 2021 Set 2 | Question: 14



Which of the following statement(s) is/are correct in the context of CPU scheduling?

- A. Turnaround time includes waiting time
- B. The goal is to only maximize CPU utilization and minimize throughput
- C. Round-robin policy can be used even when the CPU time required by each of the processes is not known apriori
- D. Implementing preemptive scheduling needs hardware support

Answer key

7.15.39 Process Scheduling: GATE CSE 2022 | Question: 32



Consider four processes P, Q, R, and S scheduled on a CPU as per round robin algorithm with a time quantum of 4 units. The processes arrive in the order P, Q, R, S, all at time $t = 0$. There is exactly one context switch from S to Q, exactly one context switch from R to Q, and exactly two context switches from Q to R. There is no context switch from S to P. Switching to a ready process after the termination of another process is also considered a context switch. Which one of the following is NOT possible as CPU burst time (in time units) of these processes?

- A. $P = 4, Q = 10, R = 6, S = 2$ B. $P = 2, Q = 9, R = 5, S = 1$
 C. $P = 4, Q = 12, R = 5, S = 4$ D. $P = 3, Q = 7, R = 7, S = 3$

gatecse-2022 operating-system process-scheduling round-robin-scheduling 2-marks

Answer key

7.15.40 Process Scheduling: GATE CSE 2023 | Question: 17



Which one or more of the following CPU scheduling algorithms can potentially cause starvation?

- A. First-in First-Out B. Round Robin
 C. Priority Scheduling D. Shortest Job First

gatecse-2023 operating-system process-scheduling multiple-selects 1-mark

Answer key

7.15.41 Process Scheduling: GATE CSE 2024 | Set 1 | Question: 15



Which of the following process state transitions is/are NOT possible?

- A. Running to Ready B. Waiting to Running
 C. Ready to Waiting D. Running to Terminated

gatecse2024-set1 operating-system process-scheduling multiple-selects

Answer key

7.15.42 Process Scheduling: GATE CSE 2024 | Set 2 | Question: 27



Consider a single processor system with four processes A, B, C, and D, represented as given below, where for each process the first value is its arrival time, and the second value is its CPU burst time.

A(0, 10), B(2, 6), C(4, 3), and D(6, 7).

Which one of the following options gives the average waiting times when preemptive Shortest Remaining Time First (SRTF) and Non-Preemptive Shortest Job First (NP-SJF) CPU scheduling algorithms are applied to the processes?

- A. $SRTF = 6, NP - SJF = 7$
 B. $SRTF = 6, NP - SJF = 7.5$
 C. $SRTF = 7, NP - SJF = 7.5$
 D. $SRTF = 7, NP - SJF = 8.5$

gatecse2024-set2 operating-system process-scheduling

Answer key

7.15.43 Process Scheduling: GATE IT 2005 | Question: 60



We wish to schedule three processes P_1 , P_2 and P_3 on a uniprocessor system. The priorities, CPU time requirements and arrival times of the processes are as shown below.

Process	Priority	CPU time required	Arrival time (hh:mm:ss)
P1	10 (highest)	20 sec	00 : 00 : 05
P2	9	10 sec	00 : 00 : 03
P3	8 (lowest)	15 sec	00 : 00 : 00

We have a choice of preemptive or non-preemptive scheduling. In preemptive scheduling, a late-arriving higher priority process can preempt a currently running process with lower priority. In non-preemptive scheduling, a late-arriving higher priority process must wait for the currently executing process to complete before it can be scheduled on the processor.

What are the turnaround times (time from arrival till completion) of P_2 using preemptive and non-preemptive scheduling respectively?

- A. 30 sec, 30 sec
 C. 42 sec, 42 sec
 B. 30 sec, 10 sec
 D. 30 sec, 42 sec

gateit-2005 operating-system process-scheduling normal

Answer key

7.15.44 Process Scheduling: GATE IT 2006 | Question: 12



In the working-set strategy, which of the following is done by the operating system to prevent thrashing?

- I. It initiates another process if there are enough extra frames.
 II. It selects a process to suspend if the sum of the sizes of the working-sets exceeds the total number of available frames.

- A. I only
 B. II only
 C. Neither I nor II
 D. Both I and II

gateit-2006 operating-system process-scheduling normal

Answer key

7.15.45 Process Scheduling: GATE IT 2006 | Question: 54



The arrival time, priority, and duration of the CPU and I/O bursts for each of three processes P_1 , P_2 and P_3 are given in the table below. Each process has a CPU burst followed by an I/O burst followed by another CPU burst. Assume that each process has its own I/O resource.

Process	Arrival Time	Priority	Burst duration (CPU)	Burst duration (I/O)	Burst duration (CPU)
P_1	0	2	1	5	3
P_2	2	3 (lowest)	3	3	1
P_3	3	1 (highest)	2	3	1

The multi-programmed operating system uses preemptive priority scheduling. What are the finish times of the processes P_1 , P_2 and P_3 ?

- A. 11, 15, 9
 B. 10, 15, 9
 C. 11, 16, 10
 D. 12, 17, 11

gateit-2006 operating-system process-scheduling normal

Answer key

7.15.46 Process Scheduling: GATE IT 2007 | Question: 26



Consider n jobs $J_1, J_2 \dots J_n$ such that job J_i has execution time t_i and a non-negative integer weight w_i .

The weighted mean completion time of the jobs is defined to be $\frac{\sum_{i=1}^n w_i T_i}{\sum_{i=1}^n w_i}$, where T_i is the completion time of job J_i . Assuming that there is only one processor available, in what order must the jobs be executed in order to minimize the weighted mean completion time of the jobs?

- A. Non-decreasing order of t_i
 C. Non-increasing order of $w_i t_i$
 B. Non-increasing order of w_i
 D. Non-increasing order of w_i/t_i

Answer key

7.15.47 Process Scheduling: GATE IT 2008 | Question: 55



If the time-slice used in the round-robin scheduling policy is more than the maximum time required to execute any process, then the policy will

- A. degenerate to shortest job first
- B. degenerate to priority scheduling
- C. degenerate to first come first serve
- D. none of the above

gateit-2008 operating-system process-scheduling easy

Answer key

7.16

Process Synchronization (52)

7.16.1 Process Synchronization: GATE CSE 1987 | Question: 1-xvi



A critical region is

- A. One which is enclosed by a pair of P and V operations on semaphores.
- B. A program segment that has not been proved bug-free.
- C. A program segment that often causes unexpected system crashes.
- D. A program segment where shared resources are accessed.

gate1987 operating-system process-synchronization

Answer key

7.16.2 Process Synchronization: GATE CSE 1987 | Question: 8a



Consider the following proposal to the "readers and writers problem."

Shared variables and semaphores:

```

aw, ar, rw, rr : integer;
mutex, reading, writing: semaphore;
initial values of variables and states of semaphores:
ar=rr=aw=rw=0
reading_value = writing_value = 0
mutex_value = 1.
Process writer;
Process reader;
begin
while true do
begin
repeat
P(mutex);
ar := ar+1;
grantread;
V(mutex);
P(reading);
read;
P(mutex);
rr := rr - 1;
ar := ar - 1;
grantwrite;
V(mutex);
other-work;
until false
end.
end.
Procedure grantread;
begin
if aw = 0
then while (rr < ar) do
begin rr := rr + 1;
V(reading)
end
end;
Procedure grantwrite;
begin
if rr = 0
then while (rw < aw) do
begin rw := rw + 1;
V(writing)
end
end

```